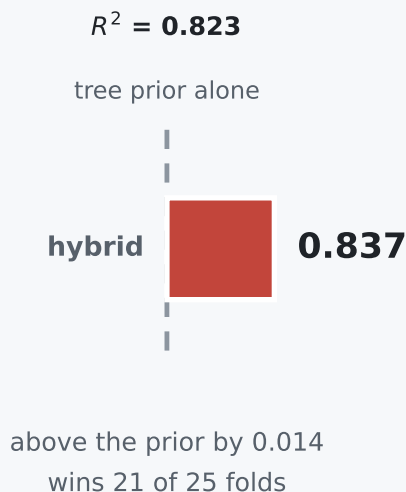


Evaluation Artefacts in Physics-Informed Band-Gap Prediction

A leakage-controlled study of 1,169 two-dimensional materials · C2DB, HSE06 band gaps · 25 fold estimates, grouped by composition

As reported

prior built on the same rows the corrector trains on



an apparent improvement — every other component held fixed

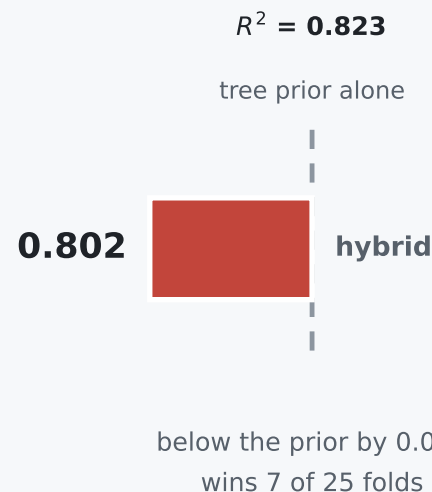
one line changed

– `fit(X).predict(X)`
+ `out_of_fold(X)`

how the tree prior
is constructed

Under leakage control

prior built out of fold, as at inference



the advantage does not survive · $p = 0.54$

What survives

stacked ensemble
of five learners

0.877

beats the prior on 25 of 25 folds · $p < 0.001$ · MAE 0.363 eV · trains in under nine seconds